

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Hussain M, Mateo J, Fizazi K, et al. Survival with olaparib in metastatic castration-resistant prostate cancer. *N Engl J Med* 2020;383:2345-57. DOI: 10.1056/NEJMoa2022485

Survival with Olaparib in Metastatic Castration-resistant Prostate Cancer

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Hussain M, Mateo J, Fizazi K, Saad F, Shore N, Sandhu S, Chi KN, Sartor O, Agarwal N, Olmos D, Thiery-Vuillemin A, Twardowski P, Roubaud G, Özgüroğlu M, Kang J, Burgents J, Gresty C, Corcoran C, Adelman CA, de Bono J

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PROFOUND INVESTIGATORS

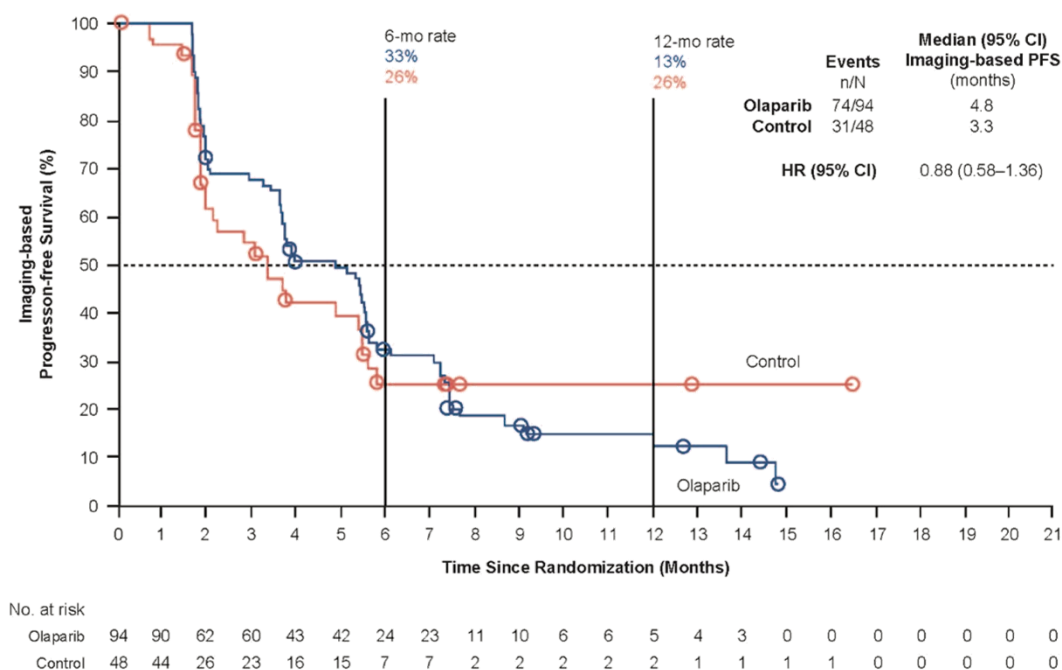
The table below lists the principal investigator for each site who participated in the study and the number of patients enrolled.

Country	Principal Investigator	Enrollment (N=387)
Argentina	Gabriela Gatica, Diego Kaen, Ernesto Korbenfeld, Luis Montes de Oca, Sandra Anabel Ostoich, Juan Sade	9
Australia	Pretoria Bilinski, Craig Gedye, Howard Gurney, Elizabeth Hovey, Edmond Kwan, Siobhan Ng, Shahneen Sandhu, Lavinia Spain, Aneta Suder, Thean Hsiang Tan	24
Austria	Richard Greil, Gero Kramer, Wolfgang Loidl, Johannes Meran, Herbert Stöger	6
Brazil	Felipe Cruz, Diogo da Rosa, Charles de Padua, Gabriel dos Anjos, Adilson Faccio, Andre Fay, Marcelo Freitas, Gustavo Giroto, Fabrício Augusto Oliveira, Suelen Martins, Marcelo Salgado, Fabio Schutz, Rodrigo Villarroel, Luis Zucca	14
Canada	Richard Casey, Kim Chi, Susan Ellard, Urban Emmenegger, Antonio Finelli, Sebastien Hotte, Nayyer Iqbal, Michael Kolinsky, Louis Lacombe, Jean-Benoit Paradis, Fred Saad, Simon Tanguay	32
Denmark	Steinbjørn Hansen	1
France	Jerome Alexandre, Emmanuelle Bompas, Karim Fizazi, Aude Flechon, Gwenaëlle Gravis, Florence Joly, Vincent Massard, Loïc Mourey, Nicolas Penel, Guilhem Roubaud, Antoine Thiery-Vuillemin, Diego Tosi, Jean-Marc Tourani	62
Germany	Manfred Binder, Thomas Ebert, Susan Feyerabend, Jochen Gleissner, Marc-Oliver Grimm, Carsten Gröllich, Eva Hellmis, Wolfgang Hölzer, Jörg Klier, Stefan Machtens, Gerald Mickisch, Günter Niegisch, Martin Schostak, Arnulf Stenzl, Henrik Suttman	12
Israel	Raanan Berger, Stephen Frank, Daniel Kejzman, Avivit Peer, Eli Rosenbaum, Avishay Sella	9
Italy	Rossana Berardi, Orazio Caffo, Ugo de Giorgi, Gaetano Facchini, Alketa Hamzaj, Emanuele Naglieri, Franco Nolè, Giuseppe Procopio, Roberto Sabbatini, Francesca Valcamonico	15
Japan	Shin Egawa, Masatoshi Eto, Yasuhisa Fujii, Kiyohide Fujimoto, Takayuki Goto, Katsuyoshi Hashine, Masashi Kato, Satoru Kawakami, Yukihiro Kondo, Nobuaki Matsubara, Ryuji Matsumoto, Masato Matsuzaki, Atsushi Mizokami, Masayoshi Nagata, Kazuo Nishimura, Chikara Ohyama, Takatsugu Okegawa, Mototsugu Oya, Hideki Sakai, Norihito Soga, Mikio Sugimoto, Hiroyoshi Suzuki, Kazuhiro Suzuki, Kenichi Tabata, Satoshi Tamada, Naoki Terada, Hiroji Uemura, Motohide Uemura, Junji Yonese, Kazuhiro Yoshimura	57
Norway	Daniel Heinrich	3
South Korea	Young Deuk Choi, Byung Ha Chung, Hong Koo Ha, Jun Hyuk Hong, Seong Soo Jeon, Jae Young Joung, Taek Won Kang, Tae Gyun Kwon, Ji Youl Lee	29
Spain	Juan Antonio Virizuela Echaburu, Carlos Álvarez Fernández, Joan Carles Galcerán, Daniel Castellano Gauna, Nuria Sala González, David Olmos Hidalgo, Javier Puente Vázquez	9
Sweden	Enrique Castellanos, Ingela Lissbrant	2
Taiwan	Chung-Hsin Chen, Po-Hui Chiang, Hsiao-Jen Chung, See-Tong Pang, Wen-Pin Su, Pai-Fu Wang, Shian-Shiang Wang, Wen-Jeng Wu, Su-Peng Yeh	15

The Netherlands	Andries Bergman, Niven Mehra, Susanne Osanto, Metin Tascilar, Robbert van Alphen, Hendrik van den Berg,	21
Turkey	Cagatay Arslan, Irfan Cicin, Mustafa Erman, Fatih Kose, Nil Mandel, Mustafa Ozguroglu, Ali Sendur, Yuksel Urun	30
United Kingdom	Johann de Bono, Paul Elliott, Stephanie Gibbs, Mark Linch, Jonathan Shamash	4
United States	Neeraj Agarwal, Andrew Armstrong, Vasileios Assikis, James Bailen, Michael Carducci, William Clark, Sarah Colonna, Gregory Daniels, Paromita Datta, Nancy Davis, Nancy Dawson, Tanya Dorff, Curtis Dunshee, Michael Goodman, Theodore Gourdin, Ivan Grunberger, Maha Hussain, Benjamin Martin, Bryan Mehlhaff, Gregory Merrick, Gregg Newman, Luke Nordquist, Yeun-Hee Park, Kalpesh Patel, Christopher Pieczonka, Daniel Saltzstein, Oliver Sartor, Stephen Savage, Peter Schlegel, Ian Schnadig, David Shaffer, Neal Shore, Kelly Stratton, Ronald Tutrone, Daniel Vaena, Nicholas Vogelzang, Jingsong Zhang	33

IMAGING-BASED PROGRESSION-FREE SURVIVAL IN COHORT B

Fig. S1. Imaging-based Progression-free Survival by Blinded Independent Central Review at the Primary Analysis Data CutOff (June 4, 2019) Among Patients with an Alteration in Genes Other than *BRCA1*, *BRCA2*, or *ATM* (Cohort B).¹

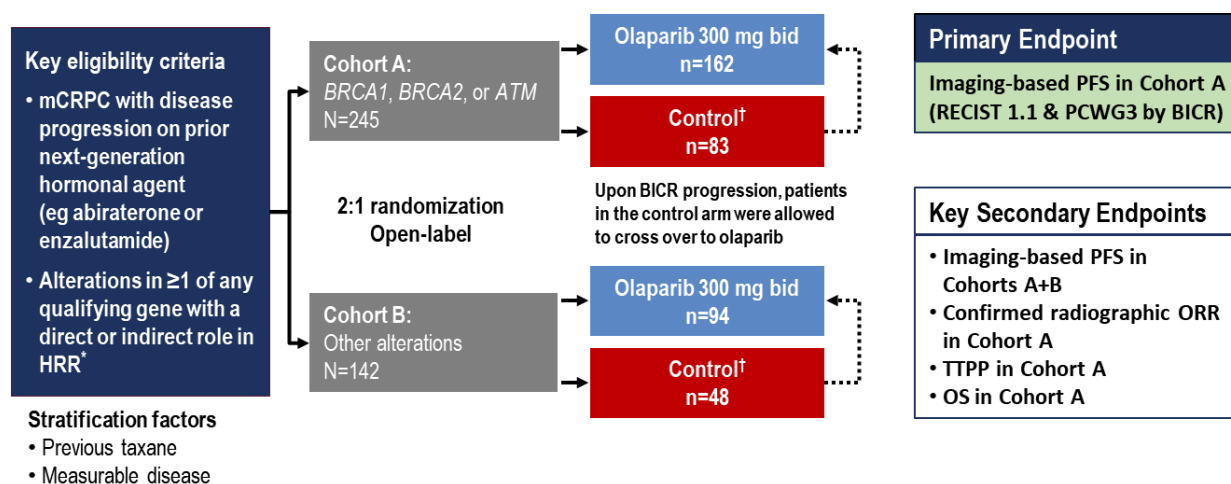


CI, confidence interval; HR, hazard ratio; mo, month; PFS, progression-free survival

SUPPLEMENTARY METHODS

Eligible patients were randomized 2:1 to receive either olaparib (300 mg twice daily) or physician's choice of either enzalutamide (160 mg twice daily) or abiraterone (1000 mg once daily, plus prednisone 5 mg twice daily; control arm). Two cohorts were enrolled: Cohort A included patients with at least one alteration in *BRCA1*, *BRCA2*, or *ATM*, irrespective of co-occurring qualifying alterations in any of the other 12 genes; Cohort B comprised patients with alterations in any of the other 12 genes (**Fig. S2**). Qualifying alterations in tumor tissue were identified using an investigational clinical trial assay, based on the FoundationOne® CDx next-generation sequencing test developed in partnership with Foundation Medicine Inc. (FMI; Cambridge, MA, USA).¹ Randomization was stratified by previous taxane use (yes/no) and Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1²)-measurable disease at baseline (yes/no). Treatment was continued until objective, blinded independent central review (BICR)-confirmed imaging-based disease progression or unacceptable toxicity.

Fig. S2. PROfound Study Design.



*An investigational Clinical Trial Assay, based on the FoundationOne® CDx next-generation sequencing test, and developed in partnership with Foundation Medicine Inc., was used to prospectively select patients harboring alterations in the following genes in their tumor tissue: *BRCA1, BRCA2, ATM, BARD1, BRIP1, CDK12, CHEK1, CHEK2, FANCL, PALB2, PPP2R2A, RAD51B, RAD51C, RAD51D, or RAD54L*.

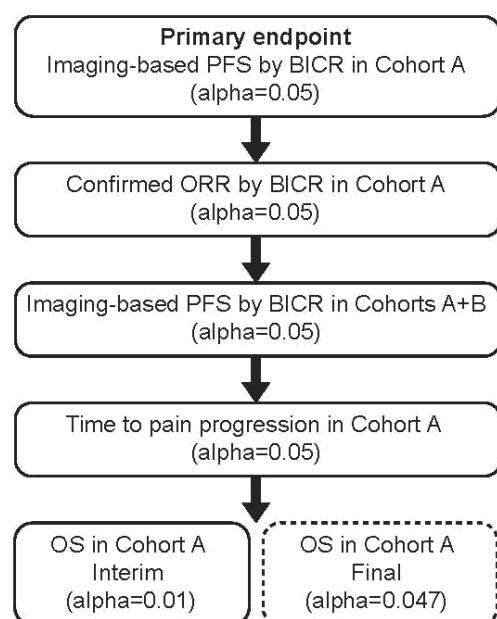
†Physician's choice of either enzalutamide (160 mg qd) or abiraterone (1000 mg qd plus prednisone [5 mg bid]).

BICR, blinded independent central review; bid, twice daily; HRR, homologous recombination repair; mCRPC, metastatic castration-resistant prostate cancer; ORR, objective response rate; OS, overall survival; PCWG3, Prostate Cancer Working Group 3 criteria; PFS, progression-free survival; qd, each day; RECIST 1.1, Response Evaluation Criteria in Solid Tumors version 1.1; TTP, time to pain progression.

Statistical analyses

Within the hierarchical multiple testing procedure used to control for the study-wide type I error rate (**Fig. S3**) OS in Cohort A was analyzed by stratified log-rank test, with the two-sided 5% alpha split at the interim analysis (0.01) and final analysis (0.047) using an O'Brien–Fleming spending function.³ For both OS and PFS2, hazard ratios (HRs) and 95% confidence intervals (CIs) were calculated using Cox proportional-hazards models with stratification factors as covariates. The Kaplan–Meier method was employed to calculate medians for each study group.

Fig. S3. Multiplicity Strategy for Primary Endpoint and Key Secondary Endpoints Maintaining Overall Type I Error Rate.



BICR, blinded independent central review; OS, overall survival.

Crossover Sensitivity Analysis

Since the study design allowed patients in the control arm the option of crossing over to receive olaparib after disease progression on control therapy (see Methods for more details), a sensitivity analysis was prespecified to explore the impact that this may have on OS. The method used for the crossover sensitivity analyses implemented rank-preserving structural failure time modeling and only adjusted for patients on the control arm who subsequently received olaparib, with a recensoring approach selected as the main OS treatment switch method. National Institute for Health and Care Excellence guidelines for switching-adjusted analyses with an active control arm were used, whereby a continuing treatment effect is assumed. Patients on the olaparib arm are always considered to be “on” treatment and control-arm patients who switch remain “on” treatment from the time of switch until death.⁴

SUPPLEMENTARY RESULTS

Patient baseline and disease characteristics

Patient baseline characteristics, including a brief summary of the distribution of the most prevalent homologous recombination repair (HRR) gene mutations, are already published,¹ but are also presented in **Table S1**.

Patient disposition

Patient disposition at the time of data cutoff for final overall survival (OS) analysis is presented in **Fig. S4**.

Table S1. Patient Characteristics at Baseline.

Characteristic	Cohort A*		Overall population*	
	Olaparib (N=162)	Control (N=83)	Olaparib (N=256)	Control (N=131)
Median age at randomization (range) — yr	68 (47–86)	67 (49–86)	69 (47–91)	69 (49–87)
Age ≥65 years at randomization — n (%)	108 (67)	60 (72)	174 (68)	97 (74)
Metastatic disease at initial diagnosis — n (%)	38 (23)	19 (23)	66 (26)	25 (19)
Missing data	7 (4)	4 (5)	11 (4)	7 (5)
Gleason score ≥ 8 — n/N (%) [†]	105/157 (67)	54/80 (67)	183/251 (73)	95/127 (75)
Patients with alteration(s) in a single gene — n (%) [‡]				
BRCA1	8 (5)	5 (6)	8 (3)	5 (4)
BRCA2	80 (49)	47 (57)	81 (32)	47 (36)
ATM	60 (37)	24 (29)	62 (24)	24 (18)
CDK12	NA	NA	61 (24)	28 (21)
Median PSA at baseline (IQR) — µg/l	62.2 (21.9, 280.4)	112.9 (34.3, 317.1)	68.2 (24.1, 294.4)	106.5 (37.2, 326.6)
Measurable disease at baseline [§] — n (%)	95 (59)	46 (55)	149 (58)	72 (55)
Metastases at baseline [§] — n (%)				
Bone only	57 (35)	23 (28)	86 (34)	38 (29)
Visceral (lung/ liver)	46 (28)	32 (39)	68 (27)	44 (34)
Other	49 (30)	23 (28)	88 (34)	41 (31)
ECOG performance status — n (%)				
0	84 (52)	34 (41)	131 (51)	55 (42)
1	67 (41)	46 (55)	112 (44)	71 (54)
2	11 (7)	3 (4)	13 (5)	4 (3)
Prior next-generation hormonal agent — n (%)				
Enzalutamide only	68 (42)	40 (48)	105 (41)	54 (41)
Abiraterone only	62 (38)	29 (35)	100 (39)	54 (41)
Abiraterone+enzalutamide	32 (20)	14 (17)	51 (20)	23 (18)
Previous taxane use — n (%)	106 (65)	52 (63)	170 (66)	84 (64)
Docetaxel only	74 (46)	32 (39)	115 (45)	58 (44)
Cabazitaxel only	2 (1)	0	3 (1)	0
Docetaxel+cabazitaxel	29 (18)	20 (24)	51 (20)	26 (20)
Paclitaxel only	1 (<1)	0	1 (<1)	0

*Cohort A included patients with at least one alteration in *BRCA1*, *BRCA2*, or *ATM*. Cohort B included patient with alterations in any of 12 other prespecified genes: *BRIP1*, *BARD1*, *CDK12*, *CHEK1*, *CHEK2*, *FANCL*, *PALB2*, *PPP2R2A*, *RAD51B*, *RAD51C*, *RAD51D*, and *RAD54L*. In both cohorts, patients in the control group received the physician's choice of enzalutamide or abiraterone.

[†]In general, scores on the Gleason scale range from 6 to 10, with higher scores indicating a worse prognosis.

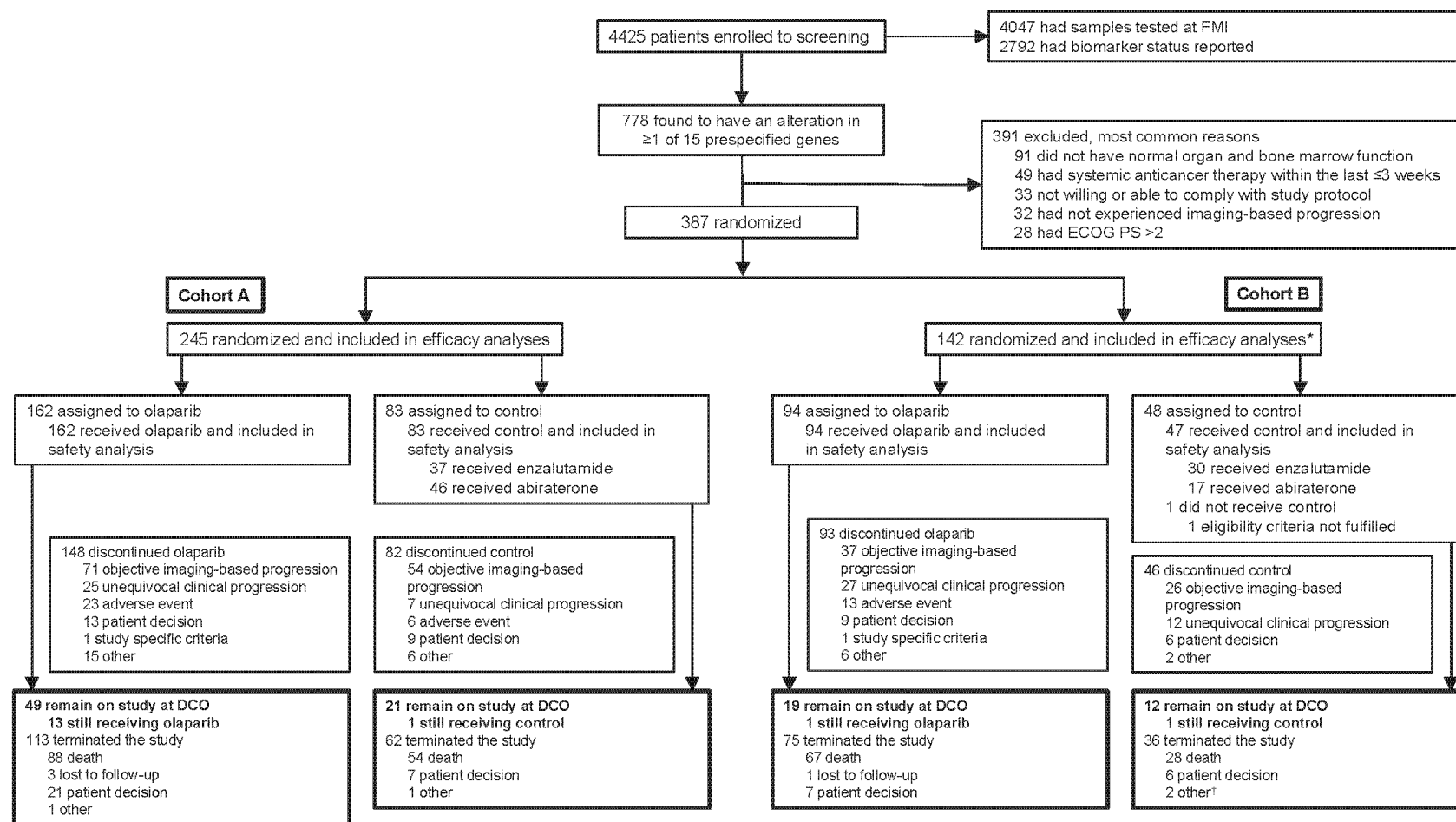
[‡]In total, 28 patients (21 in Cohort A and 7 in Cohort B) had mutations in more than one gene. Four patients were incorrectly assigned to cohort B (one in the olaparib group had an alteration in *BRCA2*, one in the control group had alterations in *BRCA2* and *CDK12*, and two in the olaparib group had alterations in *ATM*).

[§]Data were derived from electronic case-report forms as assessed by the investigator.

^{||}In total, 13 patients received a NHA for disease before a diagnosis of mCRPC; all others received a NHA after the development of mCRPC.

ECOG, Eastern Cooperative Oncology Group; IQR, interquartile range; NHA, next-generation hormonal agent; n/N, no./total no.; mCRPC, metastatic castration-resistant prostate cancer; PSA, prostate-specific antigen; yr, years.

Fig. S4. Enrollment, Screening, Randomization and Intervention in Patients to the Data Cutoff Date of the Final Overall Survival Analysis of March 20, 2020.



*Four patients were incorrectly assigned to Cohort B (one in the olaparib group had an alteration in *BRCA2*, one in the control group had alterations in *BRCA2* and *CDK12*, and two in the olaparib group had alterations in *ATM*); analyses were conducted as assigned. One patient randomized to the control arm in Cohort B did not receive treatment.

†Status was unknown for one patient.

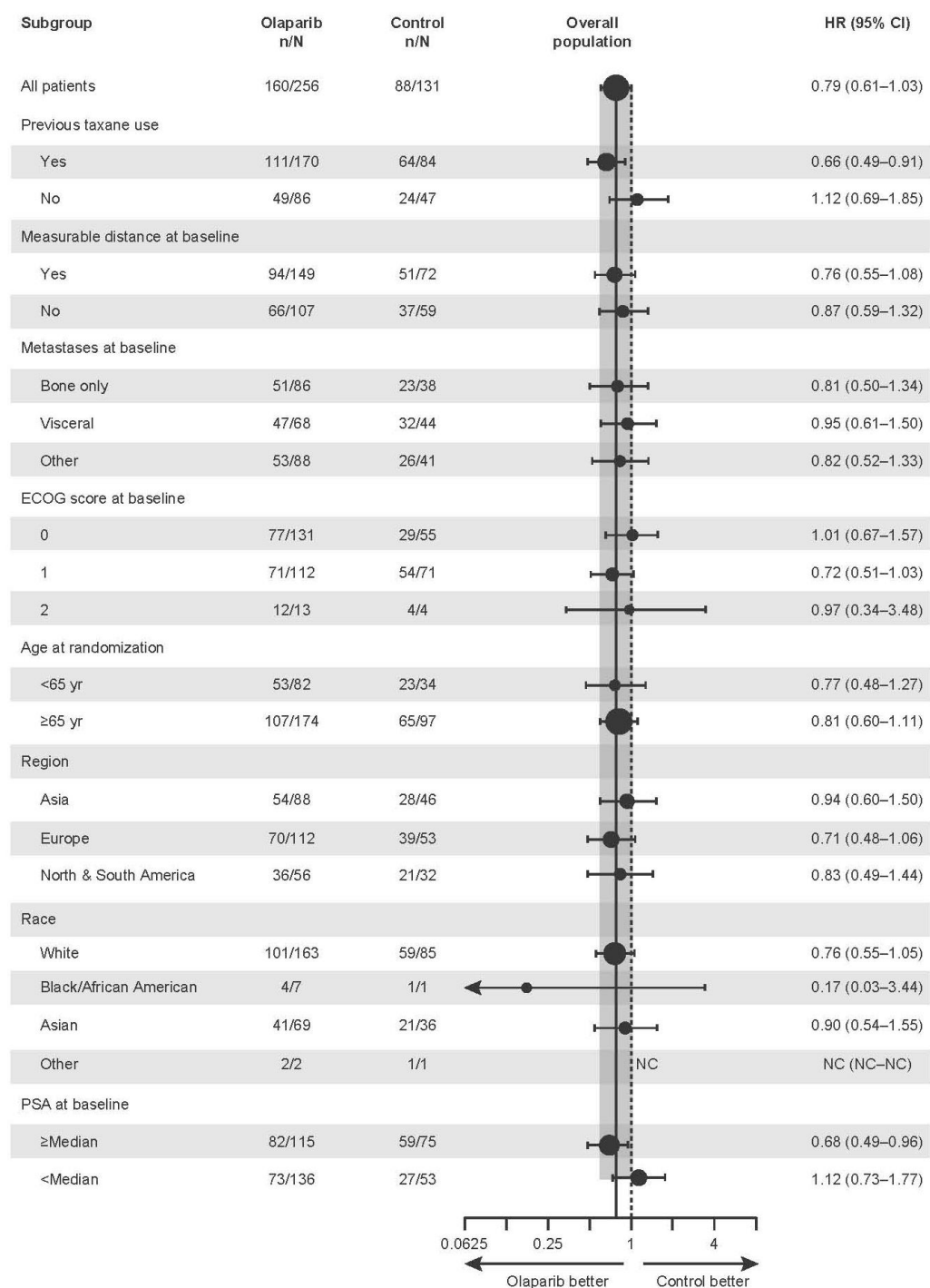
DCO, data cutoff; ECOG PS, Eastern Cooperative Oncology Group performance status.

Subgroup analyses

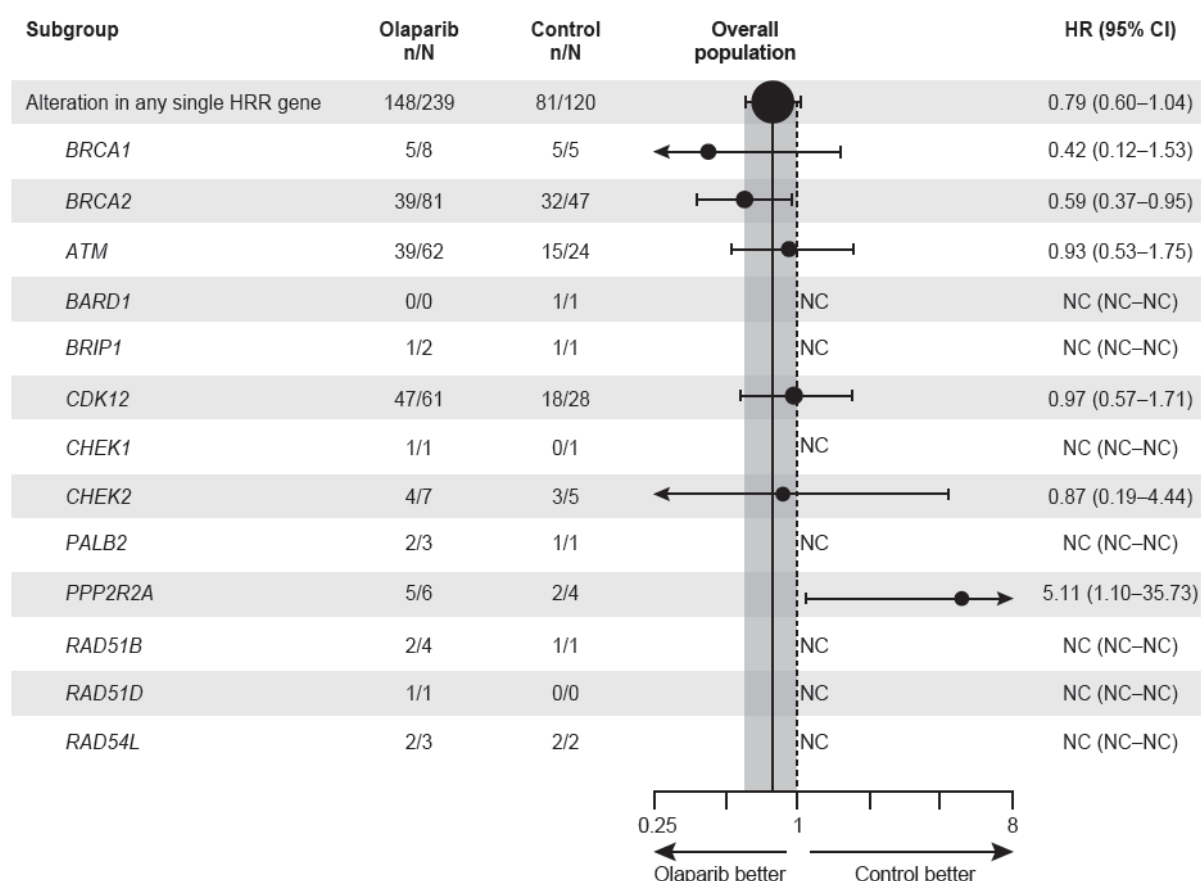
OS subgroup data for the overall population were generally consistent for treatment effect with olaparib over control (**Figs S5A**), although as with Cohort A data, the findings should be interpreted with caution given the limited number of events in some subgroups and lack of adjustment for confounding factors. Subgroup analyses of OS for patients with alterations in a single HRR gene are shown in **Figs. S5B** and **S6**. Kaplan-Meier estimates of OS are shown in **Fig. S7** together with the respective crossover sensitivity analyses for patients with alterations in non-BRCA (*BRCA1* or *BRCA2*) genes (including co-alterations), and alterations in only *ATM* or *CDK12*, and in **Fig. S8** by taxane status (with or without prior taxane treatment) for patients with an alteration only in *BRCA1* or *BRCA2*, *ATM*, or *CDK12*.

Fig. S5. Subgroup Analyses* of Overall Survival by Patient Demographic and Baseline Disease Characteristics in the Overall Population (A) and in Patients Harboring Tumors with an Alteration in a Single Homologous Recombination Repair Gene† in the Overall Population (B).

A OS by Patient Baseline Demographic and Disease Characteristics: in the Overall Population



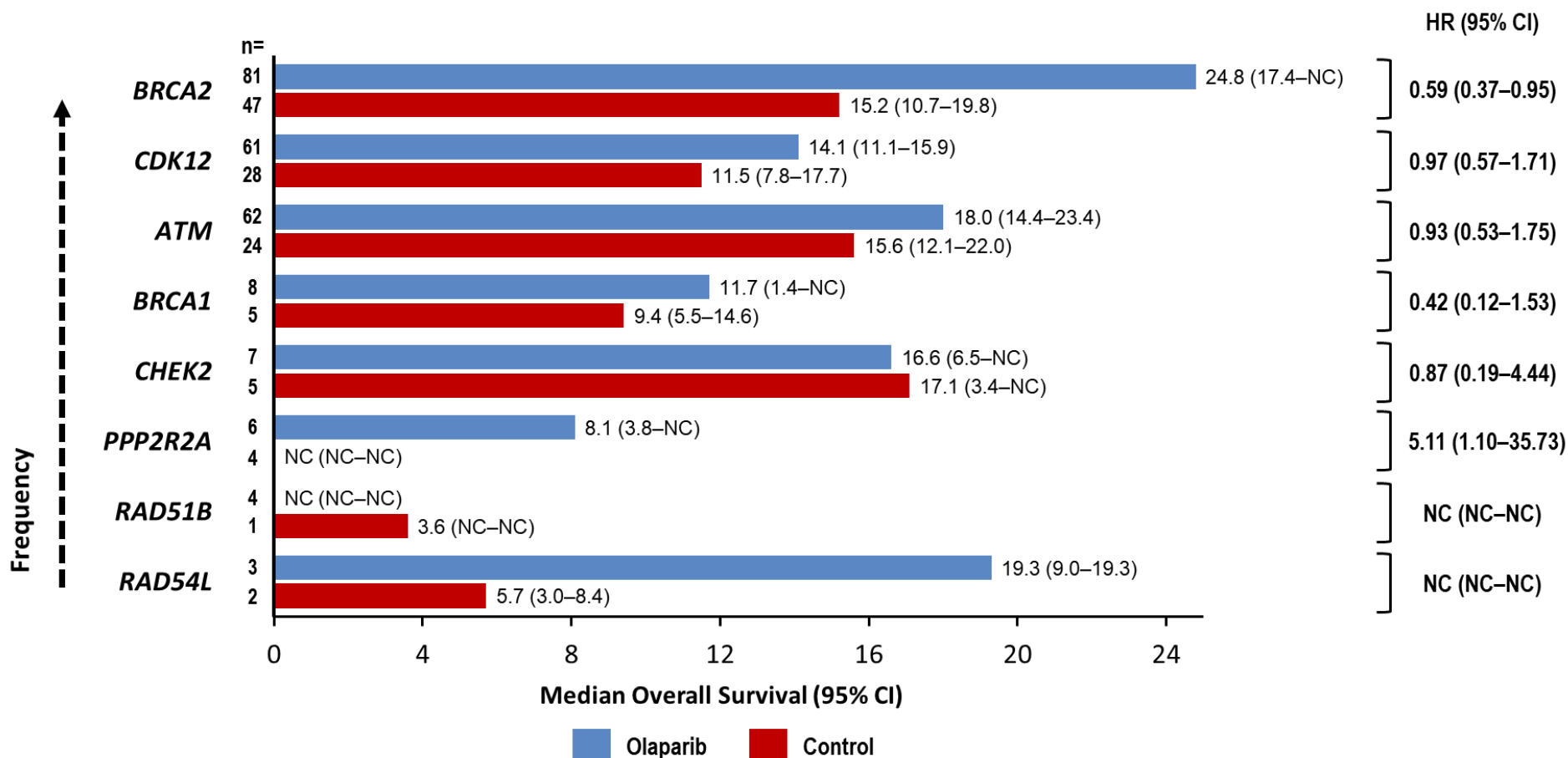
B OS in Patients with Alterations in a Single HRR[†] Gene



*Subgroups in which fewer than five survival events occurred per group were not included in the analysis. The sizes of the circles are proportional to the number of events. [†]*ATM*, *BARD1*, *BRIP1*, *CDK12*, *CHEK1*, *CHEK2*, *FANCL*, *PALB2*, *PPP2R2A*, *RAD51B*, *RAD51C*, *RAD51D*, or *RAD54L*; none of the enrolled patients harbored alterations in *FANCL* or *RAD51C*. The dashed line indicates the point of no effect (hazard ratio = 1); the solid vertical line from the point estimate for all patients has been added for reference. A hazard ratio of less than 1 favors olaparib. Note that for secondary and exploratory outcomes, which were not alpha controlled, definitive treatment effects may not be inferred.

CI, confidence interval; ECOG PS, Eastern Cooperative Oncology Group performance status; HR, hazard ratio; HRR, homologous recombination repair; NC, not calculated; n/N, number of events/number of patients in the subgroup; PSA, prostate-specific antigen.

Fig. S6. Gene-by-Gene Analysis of Overall Survival in Patients with Alterations in a Single HRR Gene. Data at the End of Each Bar are Median Overall Survival in Months (95% CI).

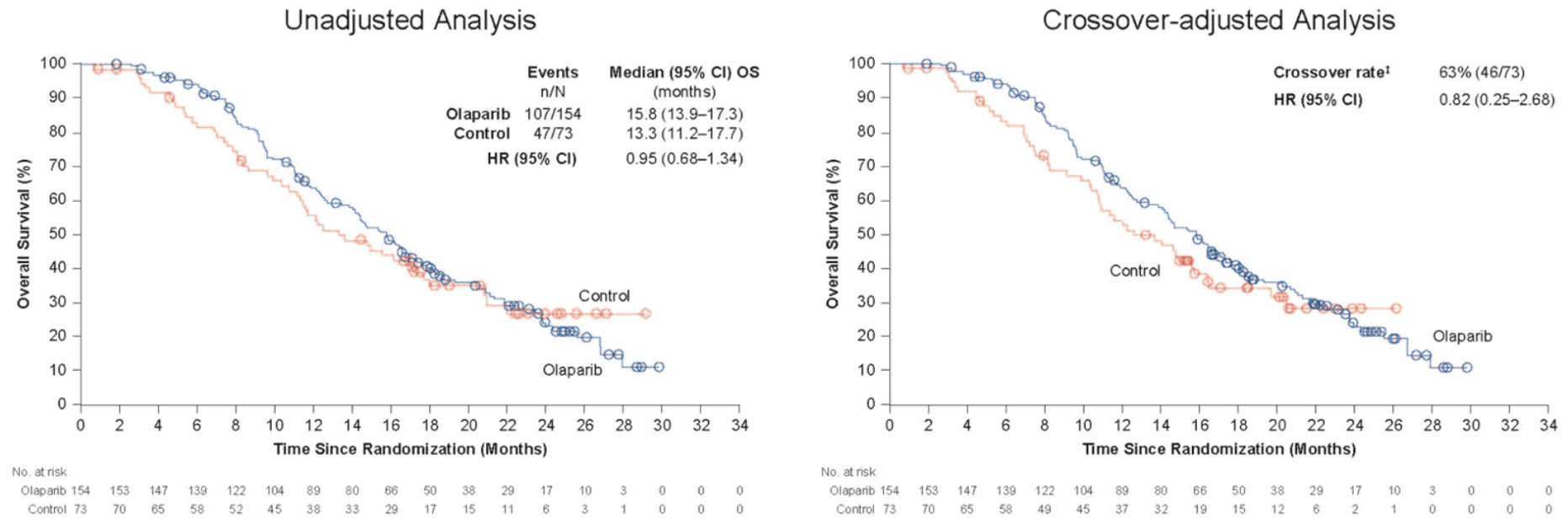


Note that for secondary and exploratory outcomes, which were not alpha controlled, definitive treatment effects may not be inferred.

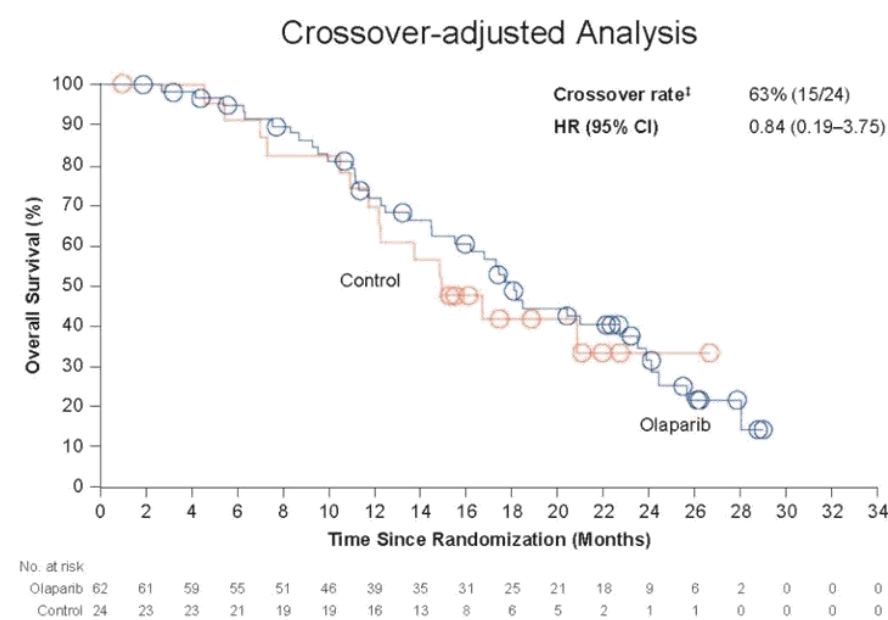
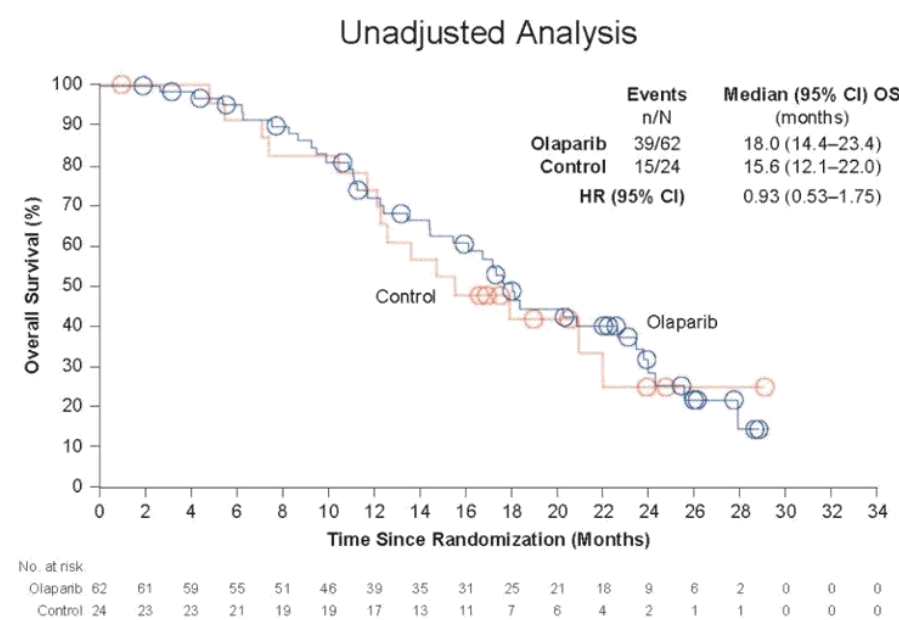
CI, confidence interval; HR, hazard ratio; HRR, homologous recombination repair; NC, not calculable; n, number of patients

Fig. S7. Kaplan–Meier Estimates of Overall Survival and Corresponding Sensitivity Crossover-Adjusted Analyses* in Patients with Alterations in Non-BRCA Genes, Including Co-alterations (A), and in Patients with Alterations in only *ATM* (B) or only *CDK12* (C).†

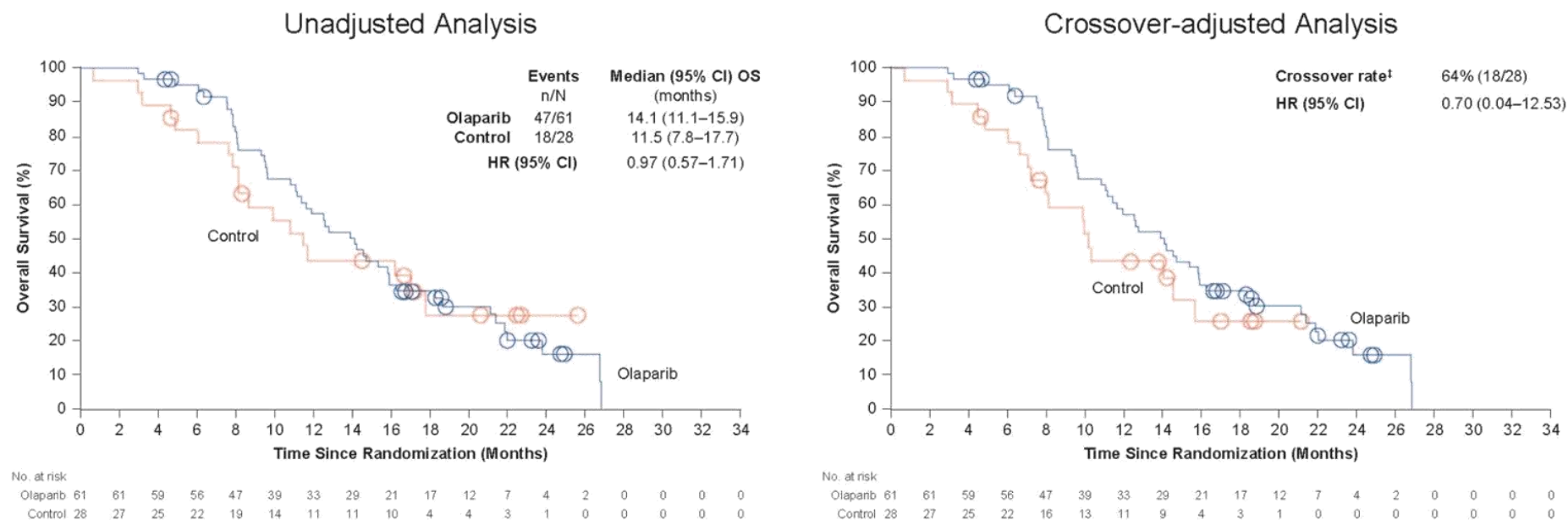
A. Kaplan–Meier Estimate of Overall Survival in Patients with Alterations in Non-BRCA Genes



B. Kaplan–Meier Estimate of Overall Survival in Patients with Alterations Only in ATM†



C. Kaplan–Meier Estimate of Overall Survival in Patients with Alterations Only in *CDK12*[†]



Note that for secondary and exploratory outcomes, which were not alpha controlled, definitive treatment effects may not be inferred.

*Re-censored; conducted using RPSFTM to demonstrate the impact on OS of crossover of patients from the control arm to receive olaparib in Cohort A and the overall population.

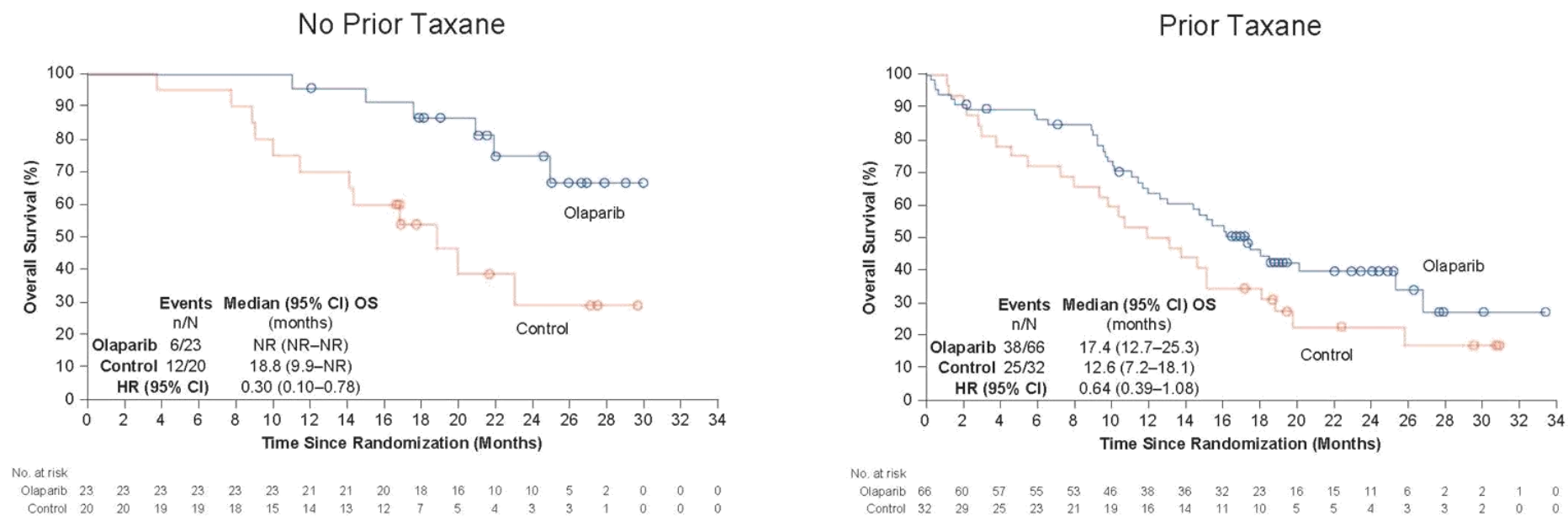
[†]Data are reported only for patients with alteration in a single gene.

[‡]Patients receiving olaparib at any time.

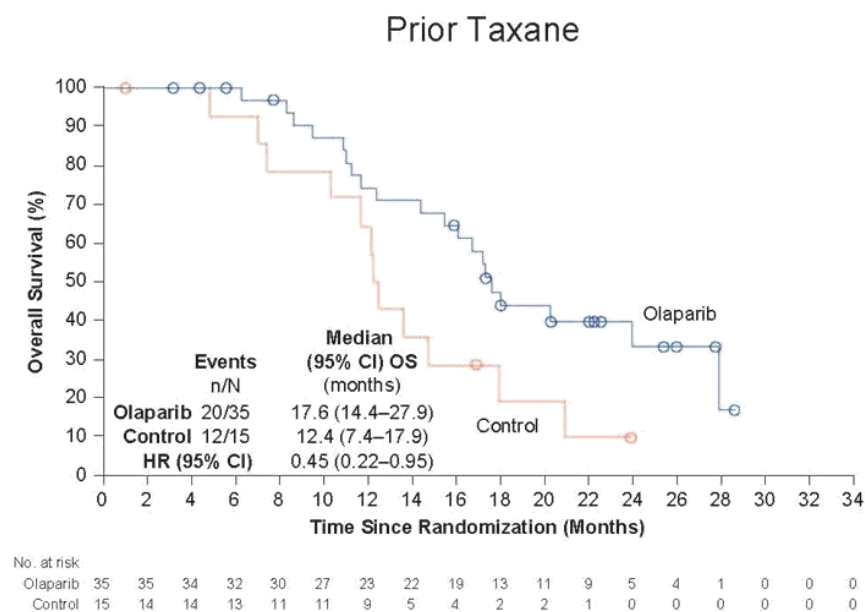
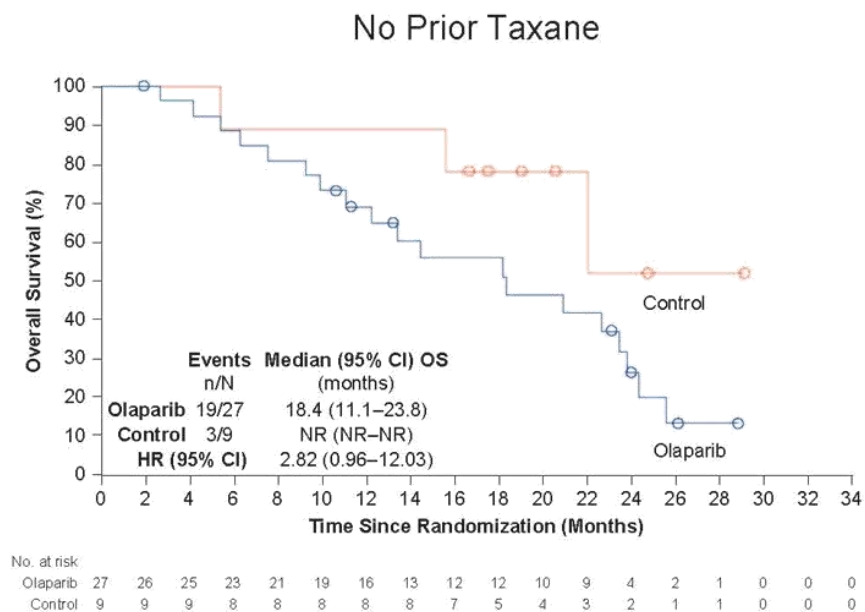
BRCA, *BRCA1/BRCA2*; CI, confidence interval; HR, hazard ratio; HRR, homologous recombination repair; n/N, number of events/number of patients; RPSFTM, rank-preserving structural failure time model.

Fig. S8. Kaplan-Meier Estimates of Overall Survival by Prior Taxane Status in Patients Harboring Alterations Only in *BRCA1* or *BRCA2* (A), *ATM* (B), or *CDK12* (C).*

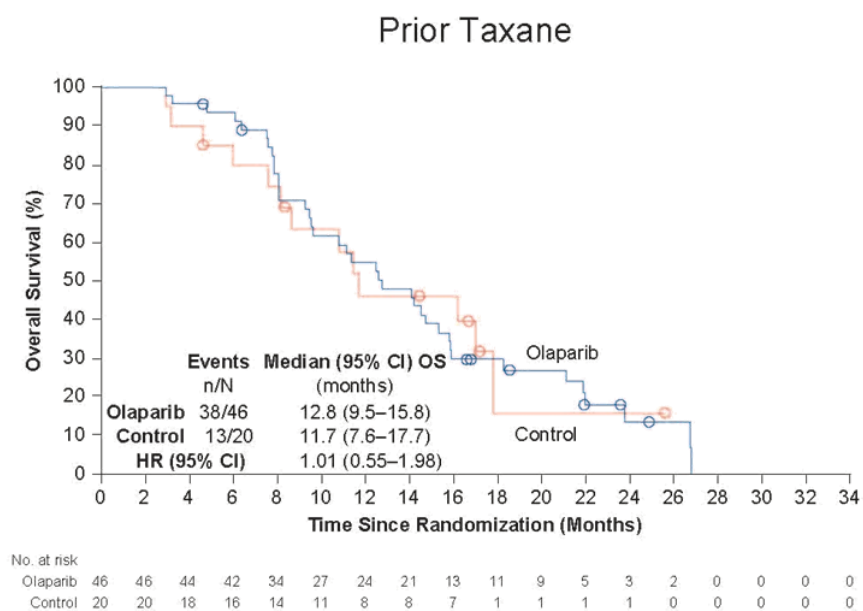
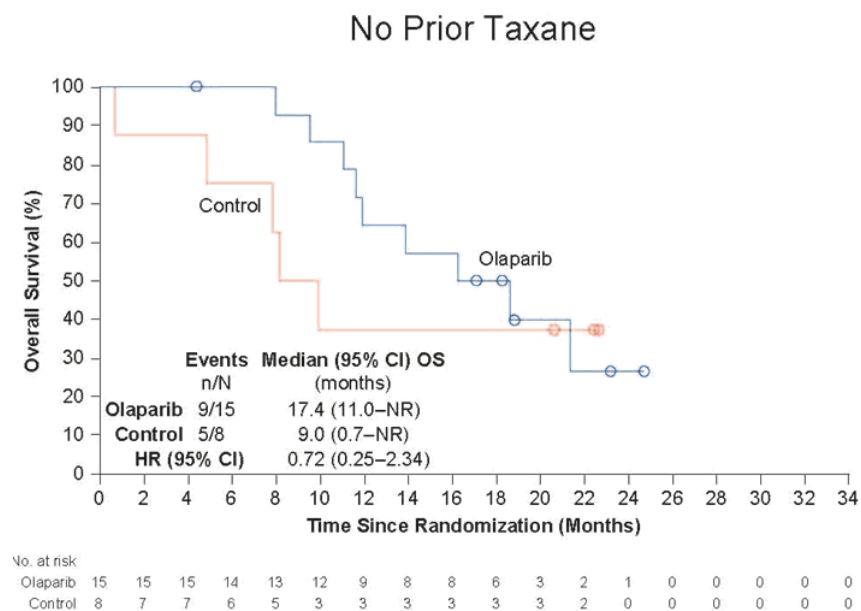
A. OS by Taxane Status: Patients with Alterations Only in *BRCA1* or *BRCA2*



B. OS by Taxane Status: Patients with Alterations Only in *ATM*



C. OS by Taxane Status: Patients with Alterations Only in *CDK12*



Note that for secondary and exploratory outcomes, which were not alpha controlled, definitive treatment effects may not be inferred.

*Data are reported only for patients with alteration in a single gene.

CI, confidence interval; HR, hazard ratio; n/N, number of events/number of patients; NR, not reached; OS, overall survival

Gene-by-gene analyses: *PPP2R2A*

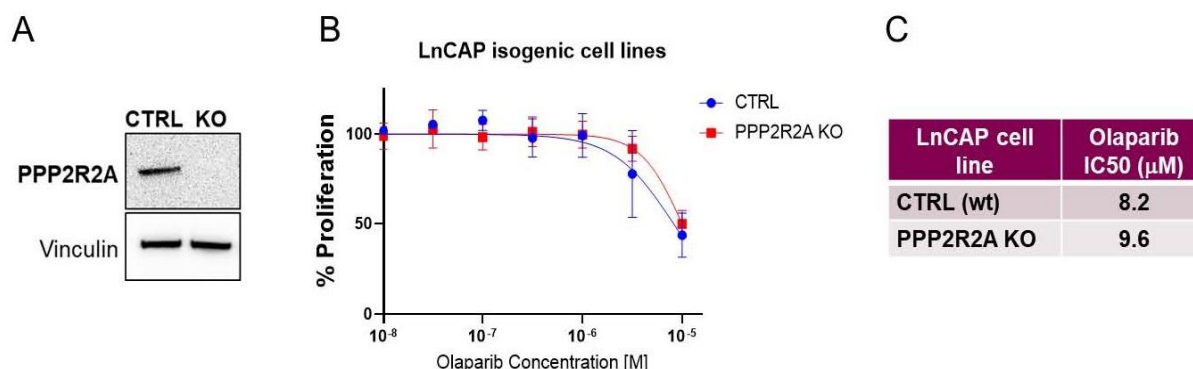
PPP2R2A is a regulatory subunit of a phosphatase of the PP2A family, which has been implicated in the negative regulation of the DNA-damage repair kinases ATM, CHEK1, and CHEK2.⁵ It has been reported that inhibition of PP2A activity impairs DNA repair.^{6,7} In particular, PPP2R2A has been demonstrated to regulate ATM phosphorylation.⁸ Based on these limited data, *PPP2R2A*, which encodes the protein, was included as one of the 15 prespecified DNA-repair genes for PROfound. PROfound data show no OS benefit for olaparib in tumors with *PPP2R2A* alterations (**Fig. S10**). No other clinical data are available regarding sensitivity to olaparib or other poly(ADP-ribose) polymerase (PARP) inhibitors in patients with a *PPP2R2A* alteration in metastatic castration-resistant prostate cancer or other cancers.

Preclinical data have now shown that olaparib is not synthetically lethal with *PPP2R2A* loss; prostate cancer cell data are presented in **Fig. S9**. Utilizing CRISPR/Cas9 technology, *PPP2R2A* knockout was generated in the prostate cancer cell line LnCaP (LnCaP-PPP2R2A-KO). Sensitivity to olaparib was tested on the LnCaP-PPP2R2A-KO cells at multiple concentrations side-by-side with LnCaP control (CTRL) cells, in a 7-day proliferation assay. As shown in **Fig. S9**, there was no enhanced sensitivity to olaparib in the LnCaP *PPP2R2A* KO cells. Additional work is ongoing to validate these findings.

Given that the preclinical data generated since the design of PROfound do not support *PPP2R2A* loss being synthetically lethal with PARP inhibition, with deleterious alterations in this gene being unlikely to confer sensitivity to olaparib or any other PARP inhibitor, an unplanned *post-hoc* sensitivity analysis was pursued to evaluate the data for the Cohort B and for the overall population excluding patients with tumors harboring *PPP2R2A* alterations (**Fig. S10**).

Fig. S9. Preclinical Data Demonstrating that *PPP2R2A*-Deficient Prostate Cancer Cells are Not Sensitive to Olaparib.

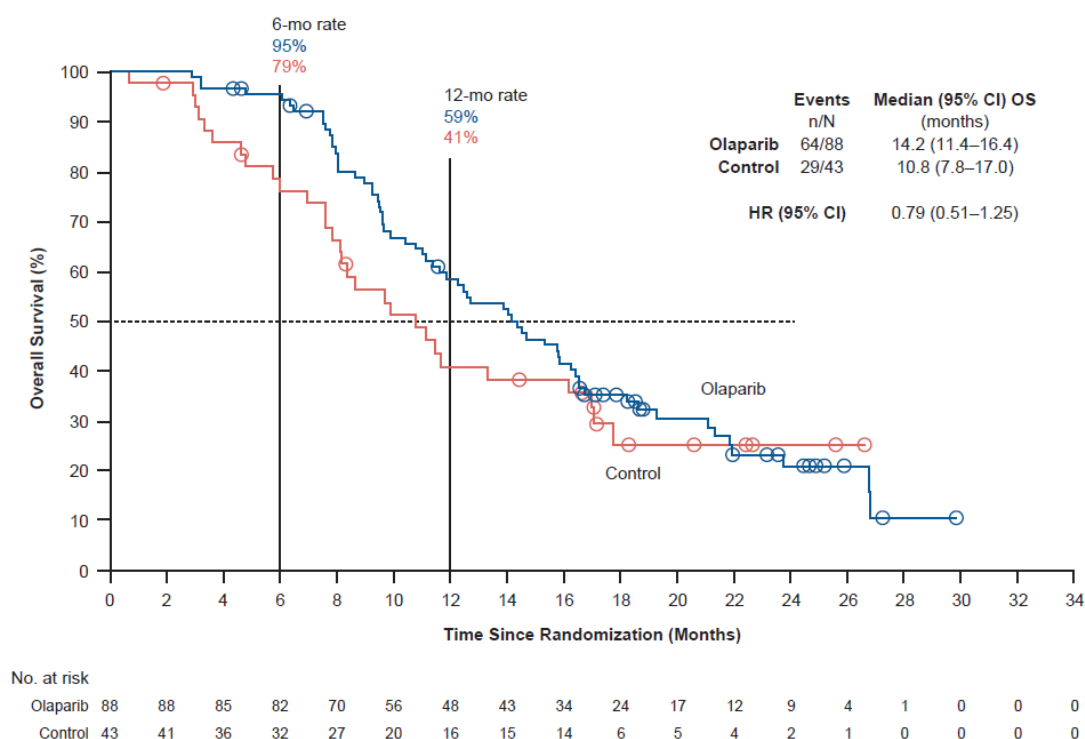
A. Western blot analysis confirmed the presence of *PPP2R2A* protein in LnCaP CTRL and absence in LnCaP *PPP2R2A* KO cells. **(B)** Cells were treated with seven different concentrations of olaparib (10–10 μ M; dilution ~1:3), and cell proliferation was evaluated by CellTiterGlo[®] luminescent cell viability assay (Promega, Leiden, Netherlands), which measures cell metabolic activity after 7 days of treatment. Proliferation was calculated as a percentage of the olaparib-treated cells versus dimethyl sulfoxide control. Each value represents the mean $\pm$ SD of three experiments. **(C)** IC₅₀ values, calculated using data from the CellTiterGlo[®] assay.



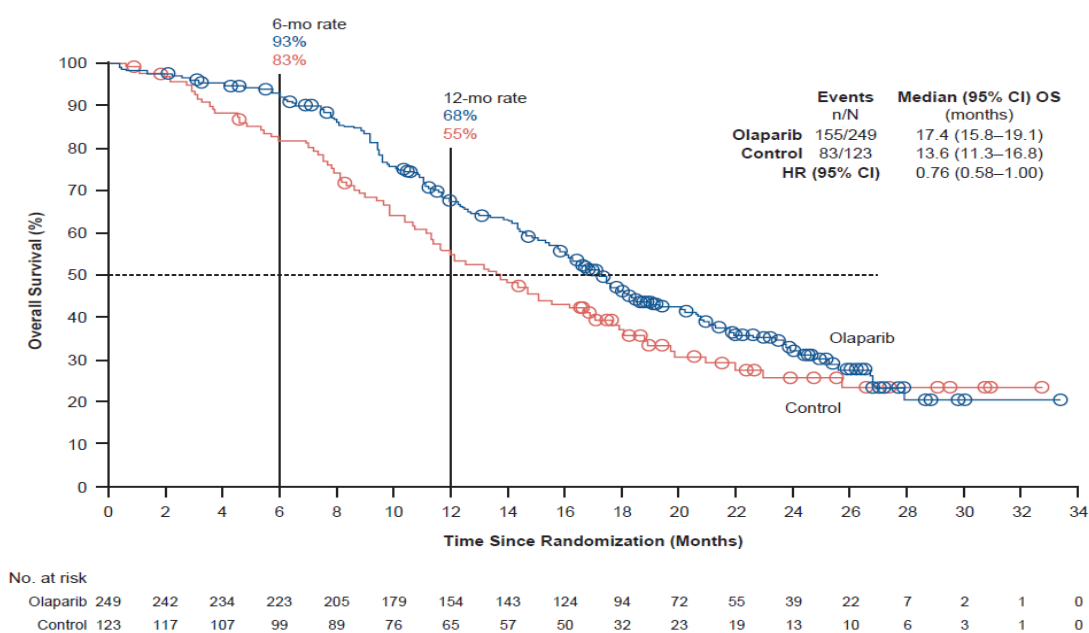
CTRL, control; DMSO, dimethyl sulfoxide; IC₅₀, half maximal inhibitory concentration; KO, knockout; SD, standard deviation.

Fig. S10 Kaplan–Meier Estimate of Overall Survival in Cohort B and in the Overall Population Excluding Patients Harboring Alterations in *PPP2R2A*.

A. K-M Estimate of OS Excluding Patients with Alterations in *PPP2R2A*: Cohort B



B. K-M Estimate of OS Excluding Patients with Alterations in *PPP2R2A*: Overall Population

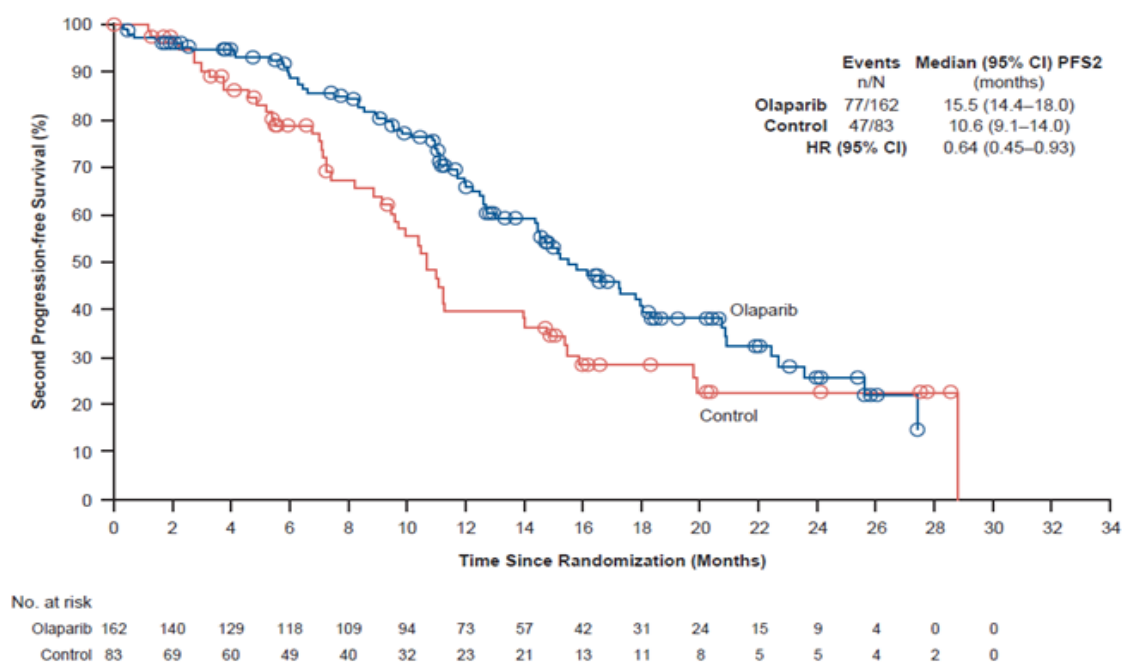


Note that for secondary and exploratory outcomes, which were not alpha controlled, definitive treatment effects may not be inferred.

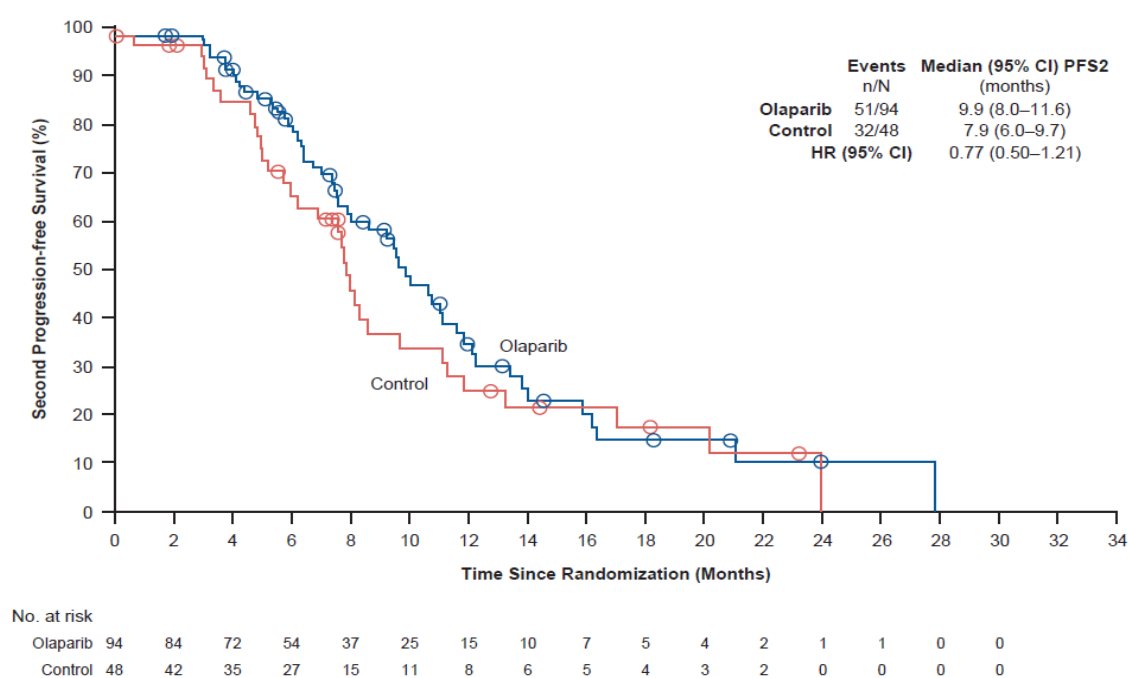
CI, confidence interval; HR, hazard ratio; mo, month; n/N, number of events/number of patients; OS, overall survival.

Fig. S11. Kaplan–Meier Estimates of Investigator-Assessed Time to Second Progression or Death in Patients with *BRCA1*, *BRCA2*, or *ATM* Alterations (A) Alterations in One of the Prespecified Genes with a Direct or Indirect Role in Homologous Recombination Repair Other than *BRCA1*, *BRCA2*, or *ATM* (B) and in the Overall Population (C).

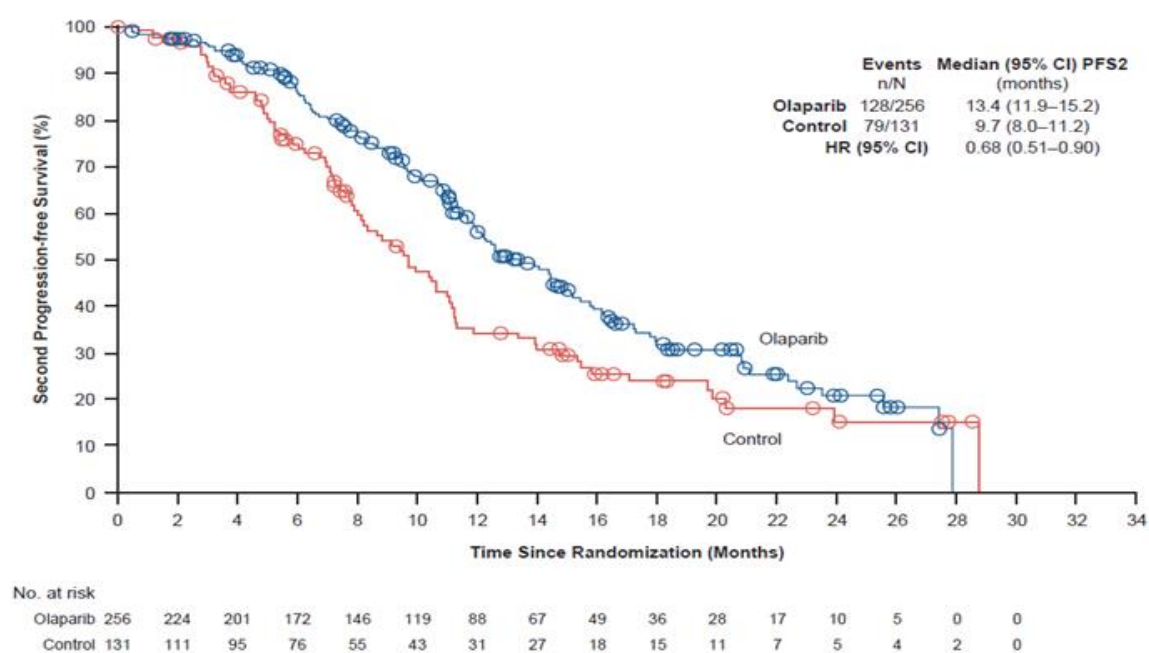
A Kaplan–Meier Estimates of PFS2: Cohort A – Patients with *BRCA1*, *BRCA2*, or *ATM* Alterations



B Kaplan–Meier Estimates of PFS2: Cohort B – Patients with Alterations in HRR Genes Other Than *BRCA1*, *BRCA2*, or *ATM*



C Kaplan–Meier Estimates of PFS2: Overall Population



CI, confidence interval; HR, hazard ratio; n/N, number of events/number of patients; PFS2, second progression-free survival or death.

Subsequent anticancer therapy

The proportion of patients who received a subsequent anticancer therapy after discontinuation of the study drug was the same in Cohort A and the overall population, at 58%. In Cohort A, treatment with subsequent PARP inhibitor therapy (olaparib) occurred in 2% of patients in the olaparib arm and in 67% of those in the control arm (**Table S2**). This disparity reflects the option offered to patients in the control arm to cross over to olaparib upon disease progression. The rate of subsequent platinum use was low.

Table S2. Subsequent Anticancer Therapies Received by >2 Patients in Cohort A and in the Overall Population Who Discontinued Either Study Treatment.

Subsequent therapy, n (%) [*]	Cohort A		Overall Population	
	Olaparib (N=162)	Control (N=83)	Olaparib (N=256)	Control (N=131)
Any	79 (49)	64 (77)	129 (50)	96 (73)
Olaparib [†]	3 (2)	56 (67)	3 (1)	86 (66)
Docetaxel	26 (16)	11 (13)	46 (18)	17 (13)
Cabazitaxel	19 (12)	10 (12)	31 (12)	15 (11)
Enzalutamide	16 (10)	1 (1)	30 (12)	1 (1)
Carboplatin	12 (7)	1 (1)	20 (8)	3 (2)
Abiraterone acetate	12 (7)	2 (2)	19 (7)	2 (2)
Lutetium (Lu-177)	6 (4)	0	10 (4)	0
Etoposide	3 (2)	0	4 (2)	0
Paclitaxel	3 (2)	0	3 (1)	0
Radium (Ra-223) dichloride	3 (2)	1 (1)	4 (2)	1 (1)
Cisplatin	2 (1)	0	4 (2)	0
Pembrolizumab	2 (1)	0	5 (2)	0

^{*}Patients with any subsequent anticancer therapy, irrespective of line. Patients could have received more than one anticancer therapy.

[†]Patients who crossed over upon disease progression with control therapy received olaparib monotherapy as a first subsequent therapy per protocol.

Safety data

The adverse-event data from the time of the primary analysis (data cutoff June 4, 2019) and the present final OS analysis (data cutoff March 20, 2020) are presented side-by-side in **Table S3** for the overall population and for the subset of patients who crossed over from the control arm to receive olaparib upon imaging-based or investigator-assessed disease progression. **Table S4** presents adverse event data from the final OS analysis separately for patients with a germline or somatic alteration in BRCA genes. Exposure-adjusted rates are provided for the most common adverse events (occurring in ≥10% of patients in either treatment arm) in **Table S5** (rates of fatigue are presented separately for grades 1, 2, and ≥3 in **Table S6**). The most frequently occurring serious adverse events were anemia (9%) and pneumonia (4%) in the olaparib arm and urinary tract infection (3%) in the control arm (**Table S7**). The rate of study treatment discontinuations was 20% (51/256) with olaparib and 8% (11/130) with control, and were attributable to the study drug in 14% (35/256) and 5% (6/130), respectively (**Table S8**).

Table S3. Summary of Adverse Events by Category at the Primary Analysis Data CutOff (June 4, 2019)¹ and the Overall Survival Data CutOff (March 20, 2020) in the Overall Population (Safety Analysis Set) and in the Subset of Those Patients With Verified Disease Progression Who Crossed Over from Control Therapy to Olaparib (Crossover).

Category, n (%)	Olaparib (N=256)		Control (N=130)*		Crossover [†]	
	PFS DCO	OS DCO	PFS DCO	OS DCO	PFS DCO (N=72)	OS DCO (N=83)
Any AE	244 (95)	246 (96)	114 (88)	115 (88)	65 (90)	77 (93)
Any AE of CTCAE ≥3	130 (51)	133 (52)	49 (38)	52 (40)	37 (51)	49 (59)
Any AE with outcome of death	10 (4)	10 (4)	5 (4)	6 (5)	2 (3)	3 (4)
Any AE attributable to study treatment	206 (81)	210 (82)	61 (47)	63 (48)	49 (68)	58 (70)
Any SAE (including with outcome of death)	91 (36)	94 (37)	36 (28)	39 (30)	17 (24)	27 (33)
Any AE leading to discontinuation of treatment	46 (18)	51 (20)	11 (8)	11 (8)	10 (14)	11 (13)

*One patient in the control arm did not receive treatment.

[†]Patients were allowed to cross over from the control arm to receive olaparib upon verification of disease progression. Safety was not assessed in the three patients in the control arm who received olaparib outside of the study.

AE, adverse event; CTCAE, Common Terminology Criteria for Adverse Events; DCO, data cutoff; OS, overall survival; PFS, progression-free survival; SAE, serious adverse event.

Table S4. Summary of Adverse Events by Category at the Final Overall Survival Data CutOff (March 20, 2020) in Patients with a Germline or Somatic Alteration in *BRCA1* or *BRCA2* (BRCaM).

Category, n (%)	Olaparib	
	Germline BRCaM (n=42)	Somatic BRCaM (n=33)
Any AE	39 (93)	33 (100)
Any AE of CTCAE ≥3	22 (52)	17 (52)
Any AE with outcome of death	0	4 (12)
Any SAE (including with outcome of death)	14 (33)	13 (39)
Any AE leading to discontinuation of treatment	8 (19)	7 (21)

AE, adverse event; CTCAE, Common Terminology Criteria for Adverse Events; SAE, serious adverse event

Table S5. Summary of Adverse Events (Irrespective of Attribution) Occurring In $\geq 10\%$ of Patients in Either Treatment Arm of the Overall Population with Exposure-Adjusted Rates.

	Overall population*			
	Olaparib (N=256)		Control (N=130) [†]	
	n (%)	Exposure-adjusted rate [‡]	n (%)	Exposure-adjusted rate [‡]
Any	246 (96)	8841	115 (89)	7134
Anemia [§]	127 (50)	978	20 (15)	346
Nausea	110 (43)	792	27 (21)	522
Fatigue or asthenia [¶]	107 (42)	795	43 (33)	920
Decreased appetite	80 (31)	500	24 (18)	441
Diarrhea	55 (21)	311	9 (7)	157
Vomiting	51 (20)	284	17 (13)	303
Constipation	49 (19)	264	19 (15)	355
Back pain	36 (14)	189	18 (14)	318
Peripheral edema	34 (13)	176	10 (8)	169
Cough	29 (11)	150	3 (2)	50
Dyspnea	27 (11)	138	5 (4)	83
Arthralgia	26 (10)	131	14 (11)	256
Urinary tract infection	21 (8)	104	15 (12)	257

*Patients who reported multiple adverse events were counted once for each type of adverse event (even if they reported multiple occurrences of a particular adverse event).

[†]One patient in the control arm did not receive treatment.

[‡]Event rate (per 1000 patient years), was calculated to account for different lengths of exposure to study treatment, for ease of comparison by arm. For example, 1 adverse event per 1000 patient-years is equivalent to one new adverse event observed among 1000 patients with exactly 1 year of exposure.

[§]The anemia category includes anemia, decreased hemoglobin level, decreased red-cell count, decreased hematocrit, erythropenia, macrocytic anemia, normochromic anemia, normochromic normocytic anemia, and normocytic anemia.

[¶]Grouped term includes fatigue and/or asthenia.

Table S6. Rates of Fatigue Separated According to Grade.

Treatment group (Cohorts A+B)	Fatigue, n (%)			
	Total	Grade 1	Grade 2	Grade ≥ 3 *
Olaparib (N=256)	69 (27)	30 (12)	35 (14)	4 (2)
Control (N=130)	28 (22)	10 (8)	15 (12)	3 (2)

*There were no reports of grade >3 fatigue in either treatment arm

Table S7 Summary of Serious Adverse Events (Irrespective of Attribution) Occurring In $\geq 2\%$ of Patients in Either Treatment Arm of the Overall Population.

	Overall population	
	Olaparib (N=256)	Control (N=130)*
	n (%)	n (%)
Any†	94 (37)	39 (30)
Anemia‡	23 (9)	0
Pneumonia	11 (4)	3 (2)
Fatigue or asthenia§	6 (2)	1 (1)
Urinary tract infection	5 (2)	4 (3)
Pulmonary embolism	5 (2)	1 (1)
Dyspnea	4 (2)	1 (1)
Thrombocytopenia	4 (2)	0
Vomiting	4 (2)	1 (1)
Sepsis	3 (1)	3 (2)
Pyrexia	3 (1)	2 (2)
Nausea	2 (1)	2 (2)
Angina pectoris	1 (<1)	2 (2)
Acute kidney injury	1 (<1)	2 (2)
Dehydration	0	3 (2)
Urinary tract obstruction	0	2 (2)
Fall	1 (<1)	2 (2)

*One patient in the control arm did not receive treatment.

†Patients who reported multiple serious adverse events were counted once for each type of adverse event (even if they reported multiple occurrences of a particular serious adverse event).

‡The anemia category includes anemia, decreased hemoglobin level, decreased red-cell count, decreased hematocrit, erythropenia, macrocytic anemia, normochromic anemia, normochromic normocytic anemia, and normocytic anemia.

§Grouped term includes fatigue and/or asthenia.

Table S8. Treatment-Related Adverse Events Leading to Study Treatment Discontinuation.

	Olaparib (N=256) n (%)	Control (N=130)* n (%)
Any[†]	35 (14)	6 (5)
Anemia [‡]	17 (7)	0
Fatigue or asthenia [§]	3 (1)	3 (2)
Neutropenia	3 (1)	0
Thrombocytopenia	3 (1)	0
Nausea	2 (1)	0
Vomiting	2 (1)	0
Pneumonia	1 (<1)	0
Pancytopenia	1 (<1)	0
Decreased appetite	1 (<1)	0
Hypocalcemia	1 (<1)	0
Insomnia	1 (<1)	0
Cerebrovascular accident	1 (<1)	0
Cough	1 (<1)	0
Interstitial lung disease	1 (<1)	0
Pneumonitis	1 (<1)	0
Diarrhea	1 (<1)	0
Acute kidney injury	1 (<1)	0
ALT increase	1 (<1)	1 (1)
Platelet count decrease	1 (<1)	0
Ischemic stroke	0	1 (1)
Pleural effusion	0	1 (1)
Myalgia	0	1 (1)
AST increase	0	1 (1)

*One patient in the control arm did not receive treatment.

[†]Patients could have experienced more than one adverse event that caused treatment discontinuation.

[‡]The anemia category includes anemia, decreased hemoglobin level, decreased red-cell count, decreased hematocrit, erythropenia, macrocytic anemia, normochromic anemia, normochromic normocytic anemia, and normocytic anemia.

[§]Grouped term includes fatigue and/or asthenia.

ALT, alanine aminotransferase; AST, aspartate aminotransferase

Pulmonary Embolism

The safety analyses conducted at the time of the primary analysis revealed an imbalance between the two treatment arms in the incidence of pulmonary embolism (4% with olaparib vs 1% with control therapy), an adverse event that is not recognized as a known complication of olaparib treatment.¹ There has since been one additional case of pulmonary embolism in the olaparib arm; the rate of pulmonary embolism with olaparib after this longer follow-up was thus similar to that reported in the primary analysis (5% vs 1%). There remains insufficient evidence to establish a causal association between pulmonary embolism and olaparib.

Adverse Events Leading to Death

Adverse events with an outcome of death (**Table S9**) were reported for 16 patients in the overall population (ten [4%] in the olaparib arm and six [5%] in the control arm). One patient in the olaparib arm had a grade 4 adverse event that was classified at the time of the primary analysis as lung infection, but was reclassified in the present final OS analysis as pneumonia and was not considered by the investigator to be causally related to the study treatment. Three patients who crossed over to olaparib from the control arm had an adverse event with an outcome of death; none were considered causally related (**Table S9**).

Table S9. Deaths Due to Any Adverse Event in the Overall Population and in Patients Who Crossed Over From Control Therapy to Olaparib Following Disease Progression.

Category, n (%) [*]	Olaparib (N=256)	Control (N=130) [†]	Crossover (N=83) [‡]
Any	10 (4)	6 (5)	3 (4)
Pneumonia	3 (1)	2 (2)	1 (1)
Cardiopulmonary failure	2 (1)	0	0
Septic shock	1 (<1)	0	0
Neutropenia/neutropenic sepsis	1 (<1)	0	1 (1)
Cardiac failure/acute cardiac failure	1 (<1)	1 (1)	0
Diverticulum intestinal	1 (<1)	0	0
Budd-Chiari syndrome	1 (<1)	0	0
Spinal stenosis	1 (<1)	0	0
Urinary retention	1 (<1)	0	0
Death/sudden death	1 (<1)	1 (1)	0
Myocardial infarction	0	0	1 (1)
Arterial thrombosis	0	1 (1)	0
Pleural effusion	0	1 (1)	0
Duodenal ulcer perforation	0	1 (1)	0

^{*}Multiple adverse events with an outcome of death were reported for two patients in the olaparib arm (spinal stenosis, urinary retention, and cardiorespiratory failure for one and neutropenia and pneumonia for the other) and one patient in the control arm (duodenal perforation and pneumonia).

[†]One patient in the control arm did not receive treatment.

[‡]Patients were allowed to cross over from the control arm to receive olaparib upon verification of disease progression. The safety analysis set for this arm did not include the three patients in the control arm who received olaparib outside of the study.

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